

# Unit 7

# Coordinate Geometry

## 1. Define coordinate plane (Cartesian plane).

The plane formed by two straight lines perpendicular to each other is called coordinate plane and the lines  $XOX'$  and  $YOY'$  are called coordinate axes.

## 2. Define order pair.

An ordered pair of real numbers  $x$  and  $y$  is a pair  $(x, y)$  in which elements are written in specific order.

**Note:**  $(x, y) \neq (y, x)$

## 3. Define origin.

The point of intersection of two coordinate axes is called origin.

## 4. Differentiate between abscissa and ordinate.

The  $x$ -coordinate of the point is called abscissa of the point  $P(x, y)$  and the  $y$ -coordinate is called its ordinate.

## 5. What are the coordinates of points lying on the axes?

- On the  $x$ -axis:  $(a, 0)$
- On the  $y$ -axis:  $(0, b)$

## 6. Define distance formula.

If  $A(x_1, y_1)$  and  $B(x_2, y_2)$  are two points and  $d$  is the distance between them, then

$$d = |AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad d \geq 0$$

**Note:**  $|AB|$  stands for  $mAB$

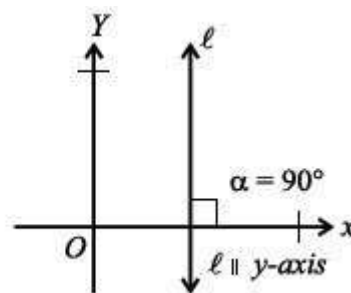
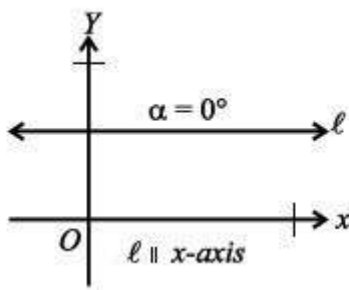
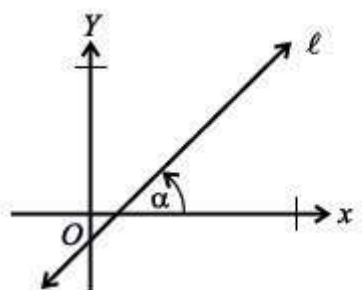
## 7. Define midpoint formula.

If  $A(x_1, y_1)$  and  $B(x_2, y_2)$  are two points in the plane, then the mid-point  $M(x, y)$  of line segment  $\overline{AB}$  is

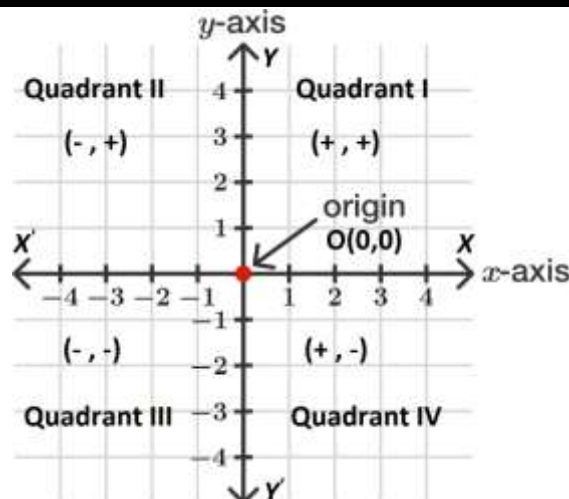
$$M(x, y) = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

## 8. What is the inclination of a line?

The inclination of a line is the angle  $\alpha$  ( $0^\circ < \alpha < 180^\circ$ ) measured counterclockwise from the positive  $x$ -axis to a non-horizontal straight line  $l$ .



- If the line  $l$  is parallel to the  $x$ -axis, then  $\alpha = 0^\circ$
- If the line  $l$  is parallel to the  $y$ -axis, then  $\alpha = 90^\circ$



Muhammad Tayyab (GHS Christian Daska)

### 9. What is the slope or gradient of a line?

Slope or gradient of an inclined path is a measure of its steepness, denoted by  $m$ . It is defined as the ratio of rise to run:

$$m = \frac{\text{rise}}{\text{run}} = \frac{y}{x} = \tan \alpha$$

In analytical geometry, for a non-vertical line with inclination  $\alpha$ ,

$$m = \tan \alpha$$



### 10. What are the special cases of slope?

- (i) If  $\ell$  is **horizontal**, its slope is **zero**.
- (ii) If  $\ell$  is **vertical**, its slope is **undefined**.
- (iii) If  $0^\circ < \alpha < 90^\circ$ , then  $m$  is **positive**.
- (iv) If  $90^\circ < \alpha < 180^\circ$ , then  $m$  is **negative**.

### 11. What is the formula for the slope of a line passing through two points?

**Theorem:** If a non-vertical line  $\ell$  with inclination  $\alpha$  passes through two points  $P(x_1, y_1)$  and  $Q(x_2, y_2)$ , then the **slope** or **gradient**  $m$  of the line is given by:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \alpha$$

**Proof:** Let  $m$  be the slope of the line  $\ell$ . Draw perpendiculars from the points:

- $\overline{PM}$  from  $P$  to the  $x$ -axis
- $\overline{QM'}$  from  $Q$  to the  $x$ -axis
- $\overline{PR}$  from  $P$  to  $\overline{QM'}$

Now:

- $m\angle RPQ = \alpha$
- $m\overline{PR} = x_2 - x_1$
- $m\overline{QR} = y_2 - y_1$

Using the definition of slope from the right triangle  $PRQ$ :

$$m = \tan \alpha = \frac{y_2 - y_1}{x_2 - x_1}$$

**Case I: When  $0 < \alpha < \frac{\pi}{2}$**

In the right triangle  $PRQ$ ,

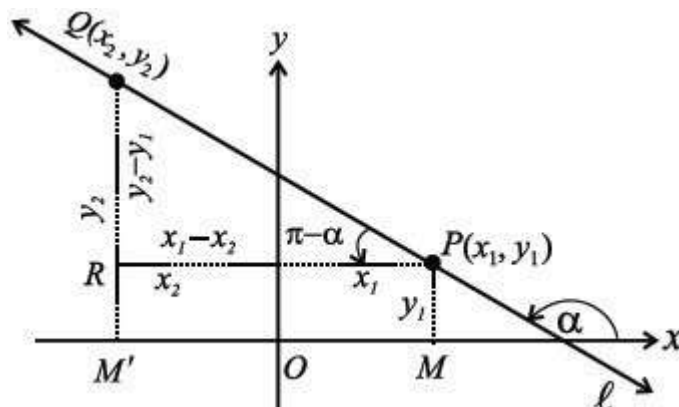
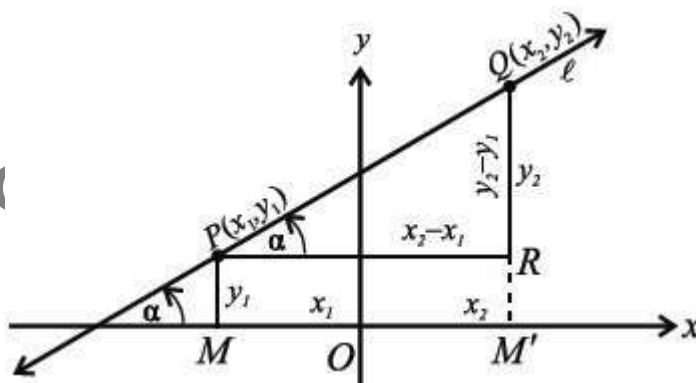
$$m = \tan \alpha = \frac{y_2 - y_1}{x_2 - x_1}$$

**Case II: When  $\frac{\pi}{2} < \alpha < \pi$**

In the right triangle  $PRQ$

$$\begin{aligned} \tan(\pi - \alpha) &= \frac{y_2 - y_1}{x_1 - x_2} \\ -\tan \alpha &= \frac{y_2 - y_1}{x_1 - x_2} \\ \tan \alpha &= \frac{y_2 - y_1}{x_2 - x_1} \\ m &= \frac{y_2 - y_1}{x_2 - x_1} \end{aligned}$$

Thus, if  $P(x_1, y_1)$  and  $Q(x_2, y_2)$  are two points on a line, then the slope of  $\overrightarrow{PQ}$  is given by:



$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

or

$$m = \frac{y_1 - y_2}{x_1 - x_2}$$

**Theorem 2:** Let two lines  $l_1$  and  $l_2$  have slopes  $m_1$  and  $m_2$ , respectively:

the symbol:

- (i)  $\parallel$  stands for "parallel".
- (ii)  $\nparallel$  stands for "not parallel".
- (iii)  $\perp$  stands for "perpendicular".

- (i) Lines are **parallel** if and only if  $m_1 = m_2$
- (ii) Lines are **perpendicular** if and only if  $m_1 = \frac{-1}{m_2}$  or  $m_1 \cdot m_2 = -1$

**Note:**

- (i)  $m \neq \frac{y_2 - y_1}{x_2 - x_1}$  or  $m \neq \frac{y_1 - y_2}{x_2 - x_1}$
- (ii) A line  $l$  is **horizontal** if and only if  $m = 0$  ( $\because \alpha = 0^\circ$ )
- (iii) A line  $l$  is **vertical** if and only if  $m$  is not defined ( $\because \alpha = 90^\circ$ )
- (iv) If  $\text{slope of } \overline{AB} = \text{slope of } \overline{BC}$ , then the points  $A, B$ , and  $C$  are collinear.

**12. Show that the points  $A(-3, 6)$ ,  $B(3, 2)$ , and  $C(6, 0)$  are collinear.**

$$\begin{aligned} \text{Slope of } \overline{AB} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{2 - 6}{3 - (-3)} \\ &= \frac{-4}{6} \\ &= \frac{-2}{3} \end{aligned}$$

$$\begin{aligned} \text{Slope of } \overline{BC} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{0 - 2}{6 - 3} \\ &= \frac{-2}{3} \end{aligned}$$

$$\therefore \text{Slope of } \overline{AB} = \text{Slope of } \overline{BC}$$

Hence, the points  $A, B$ , and  $C$  are collinear.

**13. Show that the triangle with vertices  $A(1, 1)$ ,  $B(4, 5)$ , and  $C(12, -1)$  is a right triangle.**

$$\begin{aligned} \text{Slope of } \overline{AB} = m_1 &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{5 - 1}{4 - 1} \\ &= \frac{4}{3} \end{aligned}$$

$$\begin{aligned} \text{Slope of } \overline{BC} = m_2 &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-1 - 5}{12 - 4} \\ &= \frac{-6}{8} \\ &= \frac{-3}{4} \end{aligned}$$

Muhammad Tayyab (GHS Christian Daska)

Since

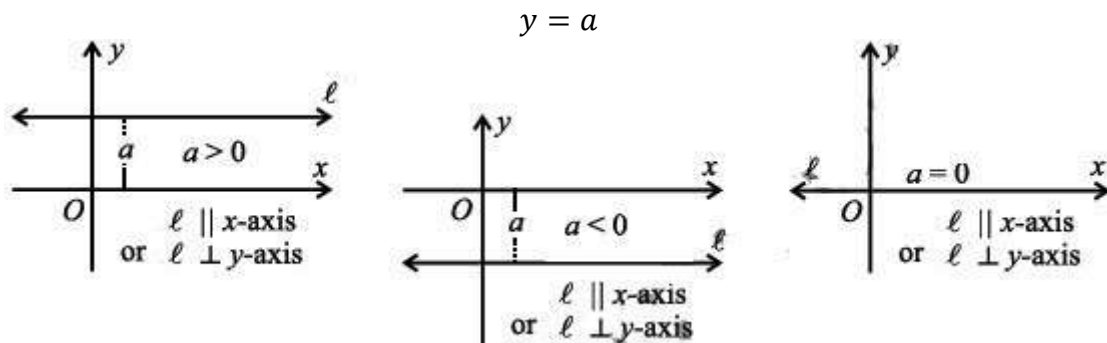
$$m_1 \cdot m_2 = \left(\frac{4}{3}\right)\left(-\frac{3}{4}\right)$$

$$m_1 \cdot m_2 = -1$$

Therefore,  $\overline{AB} \perp \overline{BC}$ . So  $\triangle ABC$  is a right triangle.

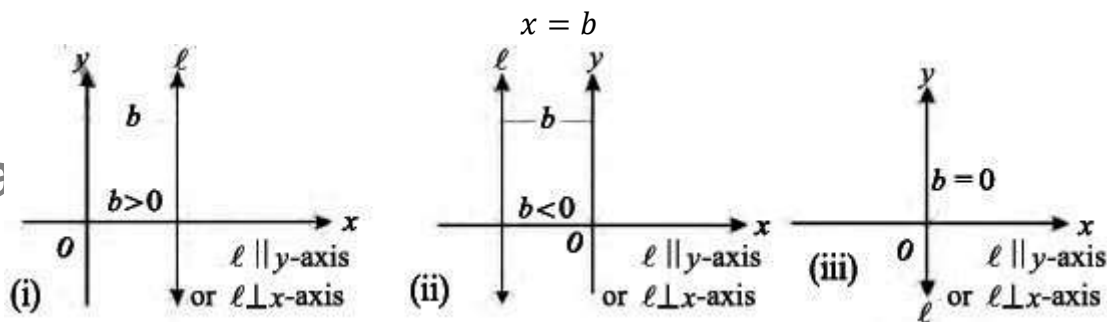
#### 14. What is the equation of a straight line parallel to the x-axis (or perpendicular to the y-axis)?

A line that goes left to right (horizontal) is called a line parallel to the x-axis. It is also said to be perpendicular to the y-axis. The equation of this line is:



#### 14. What is the equation of a straight line parallel to the y-axis (or perpendicular to the x-axis)?

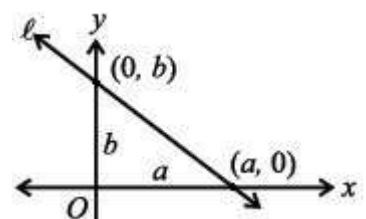
A line that goes up and down (vertical) is called a line parallel to the y-axis. It is also said to be perpendicular to the x-axis. The equation of this line is:



#### 15. What are the intercepts of a straight line?

The **x-intercept** of a line is the point where it crosses the **x-axis**. If the line crosses the x-axis at  $(a, 0)$ , then **x-intercept** =  $a$ .

The **y-intercept** of a line is the point where it crosses the **y-axis**. If the line crosses the y-axis at  $(0, b)$ , then **y-intercept** =  $b$ .



#### 16. What is the slope-intercept form of a straight line?

The equation of a non-vertical straight line with slope  $m$  and y-intercept  $c$  is:

$$y = mx + c$$

**Note:** If the line passes through the origin, then  $c = 0$ . So, the equation becomes:  $y = mx$

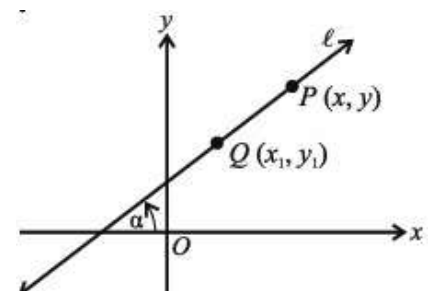
#### 17. What is the Point-Slope Form of the Equation of a Straight Line?

If a non-vertical line with slope  $m$  passes through a point  $Q(x_1, y_1)$ , then its equation is:

$$y - y_1 = m(x - x_1)$$

#### 18. What is the Symmetric Form of the Equation of a Straight Line.

If a line passes through a point  $(x_1, y_1)$  and has inclination  $\alpha$ , we know from trigonometry:



$$\tan \alpha = \frac{y - y_1}{x - x_1}$$

$$\frac{\sin \alpha}{\cos \alpha} = \frac{y - y_1}{x - x_1}$$

Rewriting:

$$\frac{x - x_1}{\cos \alpha} = \frac{y - y_1}{\sin \alpha} = r \text{ (say)}$$

This is called the **symmetric form** of the equation of a straight line.

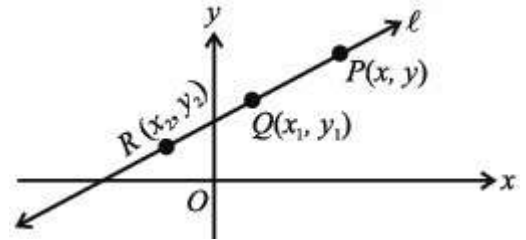
**19. What is the equation of a non-vertical straight line passing through two points  $Q(x_1, y_1)$  and  $R(x_2, y_2)$ ?**

The equation of the line passing through two given points is:

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

or

$$y - y_2 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_2)$$



**20. What is the equation of a line whose non-zero x-intercept is  $a$  and y-intercept is  $b$ ?**

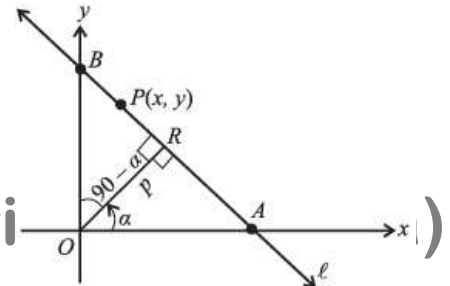
The equation of a line with non-zero  $x$ -intercept  $a$  and  $y$ -intercept  $b$  is:

$$\frac{x}{a} + \frac{y}{b} = 1$$

**21. What is the equation of a non-vertical straight line  $\ell$ , where the length of the perpendicular from the origin to  $\ell$  is  $p$ , and  $\alpha$  is the inclination of this perpendicular?**

The equation of the line is:

$$x \cos \alpha + y \sin \alpha = p$$



**Muhammad Tayyab (GHS Chri**

**22. How can the general equation of a line  $ax + by + c = 0$  be transformed into standard forms?**

The general form of a straight line  $ax + by + c = 0$  can be converted into different standard forms as follows:

**i. Slope-Intercept Form:**

$$ax + by + c = 0$$

$$by = -ax - c$$

$$y = \frac{-ax - c}{b}$$

$$y = -\frac{a}{b}x - \frac{c}{b}$$

$$y = \left(-\frac{a}{b}\right)x + \left(-\frac{c}{b}\right)$$

By comparing with  $y = mx + c_1$

$$m = -\frac{a}{b}$$

$$c_1 = -\frac{c}{b}$$

◆ How do we get the point  $\left(-\frac{c}{a}, 0\right)$ ?

From the line equation:

$$ax + by + c = 0$$

Set  $y = 0$  to find the x-intercept:

$$ax + c = 0 \Rightarrow x = -\frac{c}{a}$$

So, the point is  $\left(-\frac{c}{a}, 0\right)$

**ii. Point-Slope Form:** We already found

slope from general form:  $m = -\frac{a}{b}$

Take a known point on the line:

Let  $Q\left(-\frac{c}{a}, 0\right)$ , Using point-slope form:

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 0 &= -\frac{a}{b}\left[x - \left(-\frac{c}{a}\right)\right] \\ y &= -\frac{a}{b}x - \frac{c}{a} \end{aligned}$$

◆ How do we get the point  $\left(0, -\frac{c}{b}\right)$ ?

Set  $x = 0$  to find the y-intercept:

$$by + c = 0 \Rightarrow y = -\frac{c}{b}$$

So, the point is  $\left(0, -\frac{c}{b}\right)$

**iii. Symmetric Form:** We know from

slope:  $m = \tan \alpha = -\frac{a}{b}$

Use trigonometric identities:  $\sin \alpha = \frac{a}{\pm\sqrt{a^2+b^2}}$ ,  $\cos \alpha = \frac{b}{\pm\sqrt{a^2+b^2}}$

Using point  $Q\left(-\frac{c}{a}, 0\right)$  in  $\frac{x-x_1}{\cos \alpha} = \frac{y-y_1}{\sin \alpha} = r$  (say)

$$\frac{x + \frac{c}{a}}{\pm\sqrt{a^2+b^2}} = \frac{y - 0}{\pm\sqrt{a^2+b^2}} = r \text{ (say)}$$

**iv. Two-Point Form:** Choose two points on the line:  $A\left(-\frac{c}{a}, 0\right)$  and  $B\left(0, -\frac{c}{b}\right)$

Use two-point form:

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$y = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) + y_1$$

$$y = \frac{-\frac{c}{b} - 0}{0 + \frac{c}{a}}\left(x + \frac{c}{a}\right) + 0$$

$$y = \frac{-\frac{c}{b}}{\frac{c}{a}}\left(x + \frac{c}{a}\right)$$

$$y = \frac{-ca}{cb}\left(x + \frac{c}{a}\right)$$

$$y = \frac{-a}{b}\left(x + \frac{c}{a}\right)$$

$$y = \frac{-a}{b}x + \frac{-ac}{ba}$$

$$y = \left(-\frac{a}{b}\right)x - \frac{c}{b}$$

**v. Intercept Form:**

$$ax + by + c = 0$$

$$ax + by = -c$$

$$\frac{ax}{-c} + \frac{by}{-c} = \frac{-c}{-c}$$

$$\frac{x}{-c/a} + \frac{y}{-c/b} = 1$$

Where **x-intercept** =  $-c/a$  and **y-intercept** =  $-c/b$

**vi. Normal Form:**

$$ax + by + c = 0$$

$$\frac{ax}{\pm\sqrt{a^2+b^2}} + \frac{by}{\pm\sqrt{a^2+b^2}} = \frac{-c}{\pm\sqrt{a^2+b^2}}$$

$$x\left(\frac{a}{\pm\sqrt{a^2+b^2}}\right) + y\left(\frac{b}{\pm\sqrt{a^2+b^2}}\right) = \frac{-c}{\pm\sqrt{a^2+b^2}}$$

$$x \cos \alpha + y \sin \alpha = p$$

Where,  $\cos \alpha = \frac{a}{\pm\sqrt{a^2+b^2}}$ ,  $\sin \alpha = \frac{b}{\pm\sqrt{a^2+b^2}}$  and  $p = \frac{-c}{\pm\sqrt{a^2+b^2}}$

# Muhammad Tayyab (GHS Christian Daska)